

Confluent Digital Extrapolation and Lambert-Phase Tomography in the Fabius–Rvachev System

**Repeated spectral roots, bounded Thue–Morse filters, maximal Gamma–zeta strips,
and two-axis recovery of periodic endpoint coefficients**

Research synthesis prepared for Vladimir Reshetnikov
after a recursive audit of the complete `*.tex` corpus under
`Analysis/FabiusFunction/docs`

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Abstract

The audited PROVEIt corpus already contains the classical and formalized theory of the Fabius function and Rvachev’s up-function, exact dyadic values, inverse endpoint asymptotics, Gaussian q -binomial interpolation, Bernoulli cumulants, Bell-polynomial saddle coefficients, sharp real-axis decay of the infinite sinc product, spectral zero multiplicities, the Pascal–Rvachev hierarchy, logarithmic q -Richardson acceleration, and additive Lambert-phase locking. The recursive tree at the pinned commit contains 78 LaTeX paths, including historical revisions, imported papers, and six generated table fragments. These materials were treated as one revision-aware corpus; established results are not relabeled as discoveries here.

This report develops a different theorem package. First, for a sequence containing polynomially decorated geometric modes $n^\ell q^n$, a normalized shift polynomial with a root of multiplicity m at q annihilates all decorations with $\ell < m$. The unique minimal-degree filter, its exact condition number, its Banach-valued remainder bound, and its mixed-mode extension are obtained in closed form. The familiar Gaussian q -Richardson row is precisely the simple-root specialization.

Second, a nonminimal factorization gives a surprising stability improvement. Twisting a length- 2^m Thue–Morse block by q^{-n} produces a filter of order m whose ℓ^1 condition number is

$$\kappa_m^{\text{TM}}(q) = \frac{1 - q^{2^m}}{(1 - q) \prod_{j=0}^{m-1} (1 - q^{2^j})} \longrightarrow \frac{1}{(1 - q) \Theta_{\text{TM}}(q)}.$$

Thus the cancellation order tends to infinity while the condition number stays bounded. For $q = 1/2$ the limit is approximately 5.7112854057, whereas the minimal repeated-root filter has condition number 3^m . The price is a stencil span $2^m - 1$. The powers of two are characterized as the unique minimal-span collision-free factorization.

Third, in the exact lower-Lambert coordinate $x = \lambda 2^{-\lambda}$, every rational phase $\theta = a/d$ has multiplicative locks $\lambda_s = \rho^s(n + \theta)$ whenever $\rho \equiv 1 \pmod{d}$. Inverse powers become geometric modes and logarithmically decorated inverse powers become confluent geometric modes. This converts endpoint extraction into a stable q -Richardson problem with bounded weights, contrasting sharply with the polynomially growing condition number of the corpus’s additive locks $\lambda + j$.

Fourth, the corpus’s exact Gamma–zeta formula for the leading periodic endpoint term is shown to imply the exact maximal holomorphy strip

$$|\Im z| < \frac{\pi}{2 \log 2}.$$

Weighted Wiener-algebra bounds then make Bell-polynomial and derivative constructions quantitatively stable. Combining radial multiplicative extrapolation with a discrete Fourier transform

over rational phases yields a two-axis tomography algorithm: radial error is algebraic in the starting Lambert coordinate, while angular aliasing is exponentially small at the sharp rate $\pi^2/\log 2$.

All theorem statements are proved in the report. Claims about natural boundaries, optimal span–conditioning tradeoffs, irrational near-locks, and resurgent late-order behavior are explicitly labeled conjectural. The accompanying Python program performs exact rational cancellation checks and high-precision Gamma–zeta experiments and regenerates every table and figure.

Proof and novelty convention. “Proved here” means proved in this manuscript, not yet formalized in Lean. “New” means not found as an exact theorem or construction in the pinned 78-path corpus after path, phrase, formula, and structural searches. It is deliberately not a claim of worldwide priority. The endpoint expansion and its Gamma–zeta Fourier formula are audited inputs; the confluent filters, digital stability theorem, rational multiplicative phase geometry, maximal-strip consequence, and two-axis tomography are the new deductions.

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1 Corpus boundary and nonduplication audit

1.1 Pinned recursive inventory

The default branch changed during the audit, so all comparisons use the immutable commit printed on the title page. Its documentation subtree has recursive Git tree SHA 6063d0d90cd9a4d25ca74cd4f8b80448313fce64. The tree contains 78 LaTeX paths. GitHub’s code-search index returned fewer paths at several points because large, recent, and generated files had not all been indexed; the recursive tree, not code search, was therefore the inventory authority.

The corpus consists of current expositions, frozen revisions, a consolidated frontier volume, numerous current frontier reports, twelve Fourier-decay dossiers, source-paper transcriptions, and six small TeX fragments generated for tables or checks. Historical revisions were read as mathematical lineages: a corrected successor supersedes an older normalization, but a distinctive result surviving only in an archive file still counts as prior art. The complete path list is reproduced verbatim in [section C](#) and is also supplied as `corpus_manifest.txt` [11].

1.2 Strong negative controls

The following attractive results are already present in the corpus and are therefore excluded from the novelty claims of this report:

- the probability model for F and up, the dyadic sinc product, and the integer-zero multiplicity $1 + \nu_2(n)$;
- exact dyadic evaluation, $q = 1/2$ Pochhammer and Gaussian-binomial formulas, Lagrange interpolation, and the standard simple-root q -Richardson row;
- Prouhet cancellation and the finite Thue–Morse block product;
- Bernoulli cumulants, Bell-polynomial moment and saddle calculations, Appell deconvolution, and Barnes-type finite-prefix formulas;
- the corrected exact-Lambert endpoint coordinate, its nonconstant periodic term, all-orders inverse-power expansion, and additive phase-locked nodes $\lambda + j$;
- logarithmic Richardson acceleration of truncated sinc products and its exact Gaussian- q -binomial error series;
- the full Fourier zero divisor, spectral determinant and zeta, arithmetic rays, half-integer identities, local zero jets, and Pascal–Rvachev hierarchy;
- positive self-sampling quadrature, alias superconvergence, and finite dyadic q -Richardson recovery;
- sharp pointwise, metric, arithmetic, and envelope results for the real-axis infinite sinc product.

In particular, this report does *not* claim ordinary geometric q -Richardson extrapolation, Thue–Morse Prouhet cancellation, or Lambert phase locking separately. Its new contribution is the confluent and digitally factorized synthesis, together with the new multiplicative phase geometry and the spectral regularity consequences.

1.3 Result ledger

ID	Statement	Status
A	Shift-polynomial calculus for $n^\ell q^n$, unique minimal confluent annihilator, exact ℓ^1 condition number, and Banach-valued remainder bound.	proved here
B	General Prouhet–Richardson factorization and the twisted Thue–Morse filter with bounded infinite-order conditioning.	proved here

ID	Statement	Status
C	Minimal-span characterization of the binary exponents among collision-free subset-product factorizations.	proved here
D	Rational multiplicative Lambert phase locks and confluent cancellation of $\lambda^{-r}(\log \lambda)^\ell$ terms.	proved here
E	Exact comparison of additive and multiplicative phase-lock conditioning.	proved here
F	Exact exponential type and maximal horizontal holomorphy strip of the leading Gamma–zeta periodic endpoint term.	derived from audited corpus inputs
G	Radial q -interpolation of the periodic coefficient jet and angular DFT alias formula with a combined error theorem.	proved here
H	Weighted Wiener-algebra and Bell-polynomial bounds for certified Fourier truncation of all differential-polynomial endpoint coefficients.	proved here
I	Boundary naturality, optimal span–conditioning laws, irrational near-locks, and late-order resurgence.	conjectural/open

2 Established framework and normalization

Only the audited inputs needed later are recalled. Let F be the bounded Fabius distribution function on $[0, 1]$, and let

$$\text{up}(x) = F(1 - |x|), \quad |x| \leq 1,$$

with zero extension outside $[-1, 1]$. Under the Fourier convention used in the primary exposition,

$$\Phi(z) = \widehat{\text{up}}(z) = \prod_{h=0}^{\infty} \text{sinc}\left(\frac{\pi z}{2^h}\right), \quad \text{sinc } w = \frac{\sin w}{w}. \quad (2.1)$$

The product is entire; its nonzero integer zero at n has multiplicity $1 + \nu_2(n)$. Its logarithmic coefficients, cumulants, dyadic values, and large real-axis behavior are developed extensively in the source corpus and the relevant primary literature [1, 2, 3, 11].

For the endpoint, put

$$L = \log 2, \quad \lambda(x) = -\frac{1}{L} W_{-1}(-Lx). \quad (2.2)$$

Here W_{-1} is the lower real branch of the Lambert function [5]. Then

$$x = \lambda 2^{-\lambda}. \quad (2.3)$$

The audited all-orders expansion can be organized in the schematic exact-coordinate form

$$\mathcal{E}(\lambda) = A_0(\{\lambda\}) + \sum_{r=1}^N A_r(\{\lambda\}) \lambda^{-r} + \mathcal{O}(\lambda^{-N-1}), \quad (2.4)$$

with uniformity in the phase $\{\lambda\}$ on the real unit circle. The normalization of $\mathcal{E}$ subtracts the explicit nonperiodic main term from $\log F(x)$; its precise formula is not needed for the extrapolation algebra. The leading periodic coefficient has Fourier modes

$$\widehat{A}_0(k) = c_k = -\frac{\Gamma(-\chi_k) \zeta(1 - \chi_k)}{L}, \quad \chi_k = \frac{2\pi i k}{L}, \quad k \neq 0. \quad (2.5)$$

The mean c_0 is explicit in the corpus but plays no role in the strip and alias arguments.

We shall also allow the more general decorated expansion

$$\mathcal{E}(\lambda) = A_0(\{\lambda\}) + \sum_{r=1}^R \sum_{\ell=0}^{m_r-1} A_{r,\ell}(\{\lambda\}) \lambda^{-r} (\log \lambda)^\ell + \mathcal{R}(\lambda). \quad (2.6)$$

The current rank-one expansion is the case $m_r = 1$. The decorated form is useful for higher-rank Mellin poles, parameter derivatives, and transseries; the filter theorem below is exact whether or not a particular application ultimately contains such logarithms.

3 Confluent shift-polynomial extrapolation

3.1 The master identity

Let E denote the forward shift on sequences: $(Eu)_n = u_{n+1}$. If $W(z) = \sum_{j=0}^M w_j z^j$, set

$$W(E)u_n = \sum_{j=0}^M w_j u_{n+j}.$$

Let $D_z = z \frac{d}{dz}$ be the Euler derivative.

Lemma 3.1 (shift–Euler identity). *For $q \in \mathbb{C}^\times$, $n \in \mathbb{Z}$, and $\ell \geq 0$,*

$$W(E)[n^\ell q^n] = q^n (n + D_z)^\ell W(z) \Big|_{z=q}. \quad (3.1)$$

Proof. Since $D_z z^j = j z^j$,

$$(n + D_z)^\ell W(z) = \sum_{j=0}^M w_j (n + j)^\ell z^j.$$

Evaluation at $z = q$ and multiplication by q^n gives the claimed shifted sum. $\square$

Theorem 3.2 (confluent geometric annihilator). *Let $q_1, \dots, q_s \in \mathbb{C} \setminus \{0, 1\}$ be distinct and let $m_1, \dots, m_s \geq 1$. Define*

$$W_{\mathbf{q}, \mathbf{m}}(z) = \prod_{a=1}^s \left(\frac{z - q_a}{1 - q_a} \right)^{m_a} = \sum_{j=0}^M w_j z^j, \quad M = \sum_{a=1}^s m_a. \quad (3.2)$$

Then:

1. $W_{\mathbf{q}, \mathbf{m}}(1) = 1$, so constants are preserved;
2. for every a and every $0 \leq \ell < m_a$,

$$W_{\mathbf{q}, \mathbf{m}}(E)[n^\ell q_a^n] = 0; \quad (3.3)$$

3. among polynomials preserving constants and satisfying (3.3), this is the unique polynomial of degree at most M ;
4. if X is a Banach space and

$$u_n = u + \sum_{a=1}^s \sum_{\ell=0}^{m_a-1} v_{a,\ell} n^\ell q_a^n + r_n, \quad (3.4)$$

then

$$W_{\mathbf{q}, \mathbf{m}}(E)u_n - u = \sum_{j=0}^M w_j r_{n+j}, \quad \|W_{\mathbf{q}, \mathbf{m}}(E)u_n - u\| \leq \left(\sum_{j=0}^M |w_j| \right) \max_{0 \leq j \leq M} \|r_{n+j}\|. \quad (3.5)$$

Proof. Normalization at $z = 1$ is immediate. A zero of ordinary multiplicity m_a at q_a is also annihilated there by D_z^r for $r < m_a$, because D_z^r is a linear combination of $z^j \frac{d^j}{dz^j}$ with $j \leq r$. Expanding $(n + D_z)^\ell$ and applying theorem 3.1 proves the annihilation.

Conversely, annihilation for $\ell = 0, 1, \dots, m_a - 1$ implies successively $D_z^\ell W(q_a) = 0$, hence an ordinary zero of multiplicity at least m_a . Any admissible polynomial is therefore divisible by $\prod_a (z - q_a)^{m_a}$. Degree at most M and $W(1) = 1$ force (3.2). Substitution of (3.4) proves the exact remainder identity, and the triangle inequality gives (3.5). $\square$

3.2 Exact conditioning for positive roots

Theorem 3.3 (condition number). *If every $q_a \in (0, 1)$, then the coefficients of $W_{\mathbf{q}, \mathbf{m}}$ have strict alternating signs and*

$$\kappa_{\min}(\mathbf{q}, \mathbf{m}) := \sum_{j=0}^M |w_j| = \prod_{a=1}^s \left(\frac{1 + q_a}{1 - q_a} \right)^{m_a}. \quad (3.6)$$

Proof. The roots form a positive multiset. Expanding their product shows that the coefficient of z^j has sign $(-1)^{M-j}$. Therefore

$$\sum_j |w_j| = (-1)^M W_{\mathbf{q}, \mathbf{m}}(-1) = \prod_a \left(\frac{1+q_a}{1-q_a} \right)^{m_a}.$$

□

Example 3.4 (one repeated root). For a root q of multiplicity m ,

$$W_{q,m}(z) = \left(\frac{z-q}{1-q} \right)^m, \quad w_j = \frac{\binom{m}{j} (-q)^{m-j}}{(1-q)^m}, \quad \kappa_{\min} = \left(\frac{1+q}{1-q} \right)^m. \quad (3.7)$$

At $q = 1/2$, $m = 2$, the filter is

$$W(E)u_n = u_n - 4u_{n+1} + 4u_{n+2},$$

and it cancels both 2^{-n} and $n2^{-n}$ exactly.

Example 3.5 (mixed roots). A double root at $1/2$ and a simple root at $1/4$ give

$$W(z) = -\frac{1}{3} + \frac{8}{3}z - \frac{20}{3}z^2 + \frac{16}{3}z^3. \quad (3.8)$$

The filter cancels 2^{-n} , $n2^{-n}$, and 4^{-n} , preserves constants, and has condition number 15. The accompanying program verifies these statements in exact rational arithmetic.

3.3 The Gaussian q -Richardson row as the simple-root case

For one base $q \in (0, 1)$ and the simple roots $q, q^2, \dots, q^p$, define

$$R_{p,q}(z) = \prod_{r=1}^p \frac{z-q^r}{1-q^r} = \sum_{s=0}^p \omega_{p,s}(q) z^s. \quad (3.9)$$

The finite q -binomial theorem gives

$$\omega_{p,s}(q) = \frac{(-1)^{p-s} q^{(p-s)(p-s+1)/2}}{(q; q)_s (q; q)_{p-s}}. \quad (3.10)$$

This is the geometric Lagrange/Richardson row already developed in the corpus; for classical extrapolation context see [6]. The present theorem clarifies its spectral meaning: each inverse-power error mode corresponds to one simple root. Moreover, for $r \geq p+1$,

$$R_{p,q}(q^r) = (-1)^p q^{p(p+1)/2} \begin{bmatrix} r-1 \\ p \end{bmatrix}_q. \quad (3.11)$$

Repeated roots are therefore the natural confluent extension of the audited Gaussian q -binomial row.

4 Prouhet–Richardson factorizations and bounded digital filters

The minimal filter has the shortest possible stencil, but its conditioning is exponential in a repeated-root order. A longer factorized filter can be dramatically more stable.

4.1 A general factorized family

Theorem 4.1 (Prouhet–Richardson factorization). Fix $q \in (0, 1)$ and positive integers $a_1, \dots, a_m$. Set

$$D_{q,\mathbf{a}}(z) = \prod_{j=1}^m \frac{1-(z/q)^{a_j}}{1-q^{-a_j}}. \quad (4.1)$$

Then $D_{q,\mathbf{a}}(1) = 1$, the point $z = q$ is a zero of multiplicity m , and $D_{q,\mathbf{a}}(E)$ annihilates $n^\ell q^n$ for $0 \leq \ell < m$. Its stencil span is $A = \sum_j a_j$, and

$$\|D_{q,\mathbf{a}}\|_1 \leq \prod_{j=1}^m \frac{1+q^{a_j}}{1-q^{a_j}}. \quad (4.2)$$

Equality holds if all subset sums of the a_j are distinct.

Proof. Each factor is normalized to equal one at $z = 1$ and has a simple zero at $z = q$. The confluent annihilation follows from [theorem 3.2](#). Before like powers are collected, the absolute coefficient sum is the product of the absolute coefficient sums of the factors, namely the right-hand side of (4.2). Collecting colliding subset sums can only decrease that sum. With distinct subset sums there is no collection, hence equality. $\square$

Two extremes are already revealing. Taking every $a_j = 1$ recovers the minimal repeated root filter and its exponentially large condition number. Taking rapidly increasing a_j makes each factor in (4.2) close to one, at the cost of a very wide stencil. The binary choice is canonical because it fills its entire span without collisions.

4.2 The twisted Thue–Morse filter

Let

$$T_m(z) = \prod_{j=0}^{m-1} (1 - z^{2^j}) = \sum_{n=0}^{2^m-1} \tau_n z^n, \quad \tau_n = (-1)^{s_2(n)}. \quad (4.3)$$

Put

$$\Theta_m(q) = T_m(q) = \prod_{j=0}^{m-1} (1 - q^{2^j}), \quad \Theta_{\text{TM}}(q) = \prod_{j=0}^{\infty} (1 - q^{2^j}). \quad (4.4)$$

Theorem 4.2 (bounded Thue–Morse confluent filter). *For $q \in (0, 1)$ define*

$$\mathcal{T}_{m,q}(z) = \frac{T_m(z/q)}{T_m(1/q)} = \sum_{n=0}^{2^m-1} t_{m,q}(n) z^n. \quad (4.5)$$

Then:

1. $\mathcal{T}_{m,q}(1) = 1$ and $z = q$ is a zero of multiplicity m ;
2. the coefficients are

$$t_{m,q}(n) = \frac{(-1)^m \tau_n q^{2^m-1-n}}{\Theta_m(q)}; \quad (4.6)$$

3. for every integer N and $0 \leq \ell < m$,

$$\sum_{n=0}^{2^m-1} t_{m,q}(n) (N+n)^\ell q^{N+n} = 0; \quad (4.7)$$

4. its exact condition number is

$$\kappa_m^{\text{TM}}(q) = \sum_{n=0}^{2^m-1} |t_{m,q}(n)| \quad (4.8)$$

$$= \prod_{j=0}^{m-1} \frac{1 + q^{2^j}}{1 - q^{2^j}} = \frac{1 - q^{2^m}}{(1 - q)\Theta_m(q)}, \quad (4.9)$$

and therefore

$$\boxed{\lim_{m \rightarrow \infty} \kappa_m^{\text{TM}}(q) = \frac{1}{(1 - q)\Theta_{\text{TM}}(q)} < \infty.} \quad (4.10)$$

Proof. The first assertion follows from the product. Since

$$T_m(1/q) = (-1)^m q^{-(2^m-1)} T_m(q),$$

substitution into (4.5) gives (4.6). Multiplication by q^{N+n} removes the twist:

$$\sum_n t_{m,q}(n) (N+n)^\ell q^{N+n} = \frac{(-1)^m q^{N+2^m-1}}{\Theta_m(q)} \sum_{n=0}^{2^m-1} \tau_n (N+n)^\ell.$$

The final sum vanishes for $\ell < m$ by the standard Prouhet–Thue–Morse cancellation [4], proving (4.7) directly.

For the norm, (4.6) gives

$$\sum_n |t_{m,q}(n)| = \frac{1 + q + \dots + q^{2^m-1}}{\Theta_m(q)} = \frac{1 - q^{2^m}}{(1 - q)\Theta_m(q)}.$$

The product form uses $\prod_{j=0}^{m-1} (1 + q^{2^j}) = (1 - q^{2^m})/(1 - q)$. Finally, $\sum_j q^{2^j} < \infty$, so $\Theta_{\text{TM}}(q) > 0$ and the limit is finite. $\square$

A stability phenomenon specific to digital factorization. For $q = 1/2$, the shortest order- m filter has condition number 3^m , while the Thue–Morse filter has condition number tending to

$$\frac{2}{\Theta_{\text{TM}}(1/2)} = 5.7112854057096334466\dots$$

At order 10 the comparison is 59049 versus 5.7112854057... The digital stencil uses 1024 samples instead of 11; stability is purchased with span.

For example, at $q = 1/2$, $m = 2$,

$$\mathcal{T}_{2,1/2}(z) = \frac{1}{3} - \frac{2}{3}z - \frac{4}{3}z^2 + \frac{8}{3}z^3, \quad (4.11)$$

whose condition number is 5, compared with 9 for $1 - 4z + 4z^2$.

4.3 Why the powers of two are canonical

Theorem 4.3 (minimal span of a collision-free factorization). *Let $a_1, \dots, a_m$ be positive integers with all 2^m subset sums distinct. Then*

$$\sum_{j=1}^m a_j \geq 2^m - 1. \quad (4.12)$$

Equality holds if and only if, up to permutation,

$$(a_1, \dots, a_m) = (1, 2, 4, \dots, 2^{m-1}). \quad (4.13)$$

Proof. The distinct subset sums are 2^m distinct integers in the interval $[0, A]$, where $A = \sum_j a_j$. Hence $A+1 \geq 2^m$. If equality holds, the subset sums are exactly every integer from 0 to A . Sort the a_j . Representing 1 forces $a_1 = 1$. If $a_1, \dots, a_r$ are $1, 2, \dots, 2^{r-1}$, their subset sums fill $[0, 2^r - 1]$. Avoiding collision forces $a_{r+1} \geq 2^r$, while avoiding the gap at 2^r forces $a_{r+1} \leq 2^r$. Induction proves (4.13). $\square$

Thus the Thue–Morse filter is the unique shortest filter within the natural class whose factor expansion has no coefficient collisions and whose signed coefficients form a complete contiguous digital word. It is not globally optimal for every possible span–conditioning objective; that optimization problem is recorded in [section 11](#).

4.4 Mixed digital filters

For several modes q_a and desired confluences m_a , one may multiply the minimal filters, the digital filters, or a mixture. For instance,

$$\prod_{a=1}^s \mathcal{T}_{m_a, q_a}(z) \quad (4.14)$$

annihilates $n^\ell q_a^n$ for every $\ell < m_a$. Its condition number is bounded by the product of the individual limits

$$\prod_{a=1}^s \frac{1}{(1 - q_a)\Theta_{\text{TM}}(q_a)}.$$

This is useful when a transseries contains a small number of highly confluent modes: the sample count grows, but coefficient amplification need not.

5 Rational multiplicative Lambert phase locks

5.1 Exact phase geometry

The corpus's additive phase lock uses $\lambda, \lambda + 1, \dots$, which preserves the fractional phase for every real λ . A rational phase admits a second, multiplicative geometry.

Theorem 5.1 (multiplicative rational phase lock). *Let $\theta = a/d \in [0, 1)$ with integers $d \geq 1$ and $0 \leq a < d$. Let $\rho > 1$ be an integer satisfying*

$$\rho \equiv 1 \pmod{d}. \quad (5.1)$$

For every integer n satisfying $n + \theta \geq 1/L$ (in particular, every $n \geq 2$), define

$$\lambda_s = \rho^s(n + \theta), \quad x_s = \lambda_s 2^{-\lambda_s}, \quad s \geq 0. \quad (5.2)$$

Then $\lambda(x_s) = \lambda_s$ on the lower branch, and $\{\lambda_s\} = \theta$ for every s . With $q = \rho^{-1}$ and $H = (n + \theta)^{-1}$,

$$\lambda_s^{-r} (\log \lambda_s)^\ell = H^r q^{rs} (\log(n + \theta) + s \log \rho)^\ell. \quad (5.3)$$

Thus a logarithmically decorated inverse power becomes a polynomially decorated geometric mode in the orbit index s .

Proof. Because $d \mid (\rho^s - 1)$,

$$\rho^s \left(n + \frac{a}{d} \right) = \rho^s n + \frac{(\rho^s - 1)a}{d} + \frac{a}{d},$$

and the first two terms are integers. Moreover, $\lambda_s \geq \lambda_0 = n + \theta \geq 1/L$, so $-L\lambda_s \leq -1$ and the identity $(-L\lambda_s) \exp(-L\lambda_s) = -Lx_s$ is inverted by W_{-1} , giving $\lambda(x_s) = \lambda_s$. Formula (5.3) is direct. $\square$

5.2 Confluent endpoint extraction

Theorem 5.2 (Lambert–confluent Richardson filter). *Assume the phase-locked expansion (2.6) along the orbit in theorem 5.1. Define*

$$W(z) = \prod_{r=1}^R \left(\frac{z - q^r}{1 - q^r} \right)^{m_r}. \quad (5.4)$$

Then $W(E_s)$, applied to the samples $\mathcal{E}(\lambda_s)$, preserves $A_0(\theta)$ and cancels every displayed decorated term in (2.6). If $q \in (0, 1)$, its exact weight norm is

$$\kappa = \prod_{r=1}^R \left(\frac{1 + q^r}{1 - q^r} \right)^{m_r}. \quad (5.5)$$

The minimal factors in (5.4) may be replaced by the digital factors $\mathcal{T}_{m_r, q^r}$ to reduce high-confluence coefficient growth.

Proof. By (5.3), the term with indices (r, ℓ) is a linear combination of $s^j q^{rs}$ for $0 \leq j \leq \ell < m_r$. Apply theorems 3.2 and 3.3. Digital replacement follows from theorem 4.2. $\square$

For the current pure inverse-power expansion, take $m_r = 1$ for $1 \leq r \leq p$. The weights are exactly (3.10), now acting along a rational multiplicative Lambert orbit. If the expansion is available through A_{p+1} , then

$$\sum_{s=0}^p \omega_{p,s}(q) \mathcal{E}(\lambda_s) = A_0(\theta) + (-1)^p q^{p(p+1)/2} \frac{A_{p+1}(\theta)}{(n + \theta)^{p+1}} + \mathcal{O}(n^{-p-2}). \quad (5.6)$$

5.3 Conditioning versus the additive lock

The additive construction samples $\lambda + j$ and interpolates in $h_j = (\lambda + j)^{-1}$. Evaluating the degree- p interpolant at $h = 0$ gives exact weights

$$a_{p,j}(\lambda) = \frac{(-1)^{p-j} (\lambda + j)^p}{j!(p-j)!}, \quad 0 \leq j \leq p. \quad (5.7)$$

Consequently

$$\kappa_{\text{add}}(\lambda, p) = \sum_{j=0}^p \frac{(\lambda + j)^p}{j!(p-j)!} = \frac{2^p}{p!} \lambda^p + \mathcal{O}_p(\lambda^{p-1}). \quad (5.8)$$

By contrast, the multiplicative pure-power row has

$$\kappa_{\text{mult}}(q, p) = \prod_{r=1}^p \frac{1+q^r}{1-q^r} \longrightarrow \frac{(-q; q)_{\infty}}{(q; q)_{\infty}} < \infty \quad (p \rightarrow \infty), \quad (5.9)$$

independently of the starting n .

Numerical conditioning comparison. For phase $1/8$, choose $\rho = 9$ and $q = 1/9$. At $\lambda = 30 + 1/8$ and order $p = 10$, the exact formulas give

$$\kappa_{\text{add}} = 8.82199160936518 \times 10^{11}, \quad \kappa_{\text{mult}} = 1.28521058629968.$$

This compares linear weight amplification only. Multiplicative samples correspond to extremely separated endpoint arguments $x_s = \lambda_s 2^{-\lambda_s}$ and therefore demand wide dynamic range or logarithmic evaluation; they are stable but not computationally free.

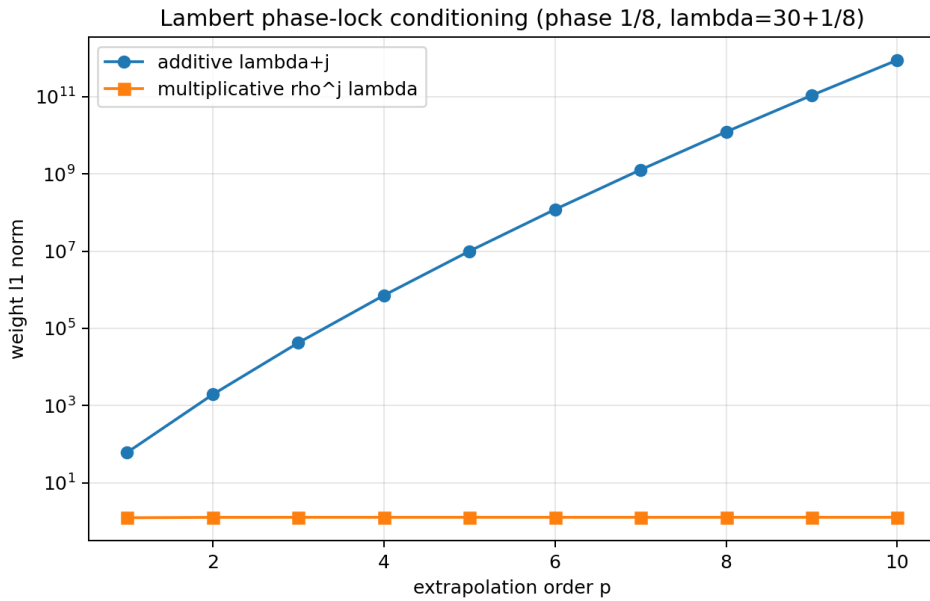


Figure 1: Exact ℓ^1 norms for additive and multiplicative phase-locked extrapolation. The multiplicative row uses the rational phase $1/8$ and $\rho = 9$.

6 The maximal Gamma–zeta strip

The exact mode formula (2.5) contains more analytic information than mere smoothness of the periodic term.

6.1 Exact exponential type of the Fourier coefficients

For $t > 0$, the classical Gamma identity [10]

$$|\Gamma(it)|^2 = \frac{\pi}{t \sinh(\pi t)} \quad (6.1)$$

implies

$$\log |\Gamma(it)| = -\frac{\pi t}{2} - \frac{1}{2} \log t + \frac{1}{2} \log(2\pi) + o(1). \quad (6.2)$$

Classical bounds on the line $\Re s = 1$ [7, 8, 9] give

$$|\zeta(1+it)| \ll \log(2+t), \quad |\zeta(1+it)|^{-1} \ll \log(2+t), \quad (6.3)$$

so $\log |\zeta(1+it)| = o(t)$.

Theorem 6.1 (exact Fourier exponential type). *Let c_k be given by (2.5). Then*

$$\boxed{\lim_{|k| \rightarrow \infty} -\frac{1}{|k|} \log |c_k| = \alpha := \frac{\pi^2}{\log 2}.} \quad (6.4)$$

More quantitatively,

$$|c_k| \ll \frac{\log(2+|k|)}{\sqrt{|k|}} \exp\left(-\frac{\pi^2}{\log 2} |k|\right). \quad (6.5)$$

Proof. For $k > 0$, put $t_k = 2\pi k/L$. Equations (2.5), (6.2), and (6.3) give

$$\log |c_k| = -\frac{\pi t_k}{2} + o(t_k) = -\frac{\pi^2}{L} k + o(k).$$

The $k < 0$ case follows by conjugation. Keeping the elementary prefactor in (6.1) and the upper zeta bound gives (6.5). $\square$

6.2 Maximal holomorphy

Theorem 6.2 (maximal horizontal strip). *The one-periodic function*

$$A_0(z) = c_0 + \sum_{k \neq 0} c_k e^{2\pi i k z} \quad (6.6)$$

extends holomorphically to

$$\boxed{\mathcal{S}_* = \{z \in \mathbb{C} : |\Im z| < \sigma_*\}, \quad \sigma_* := \frac{\pi}{2 \log 2}.} \quad (6.7)$$

It has no holomorphic continuation to any strictly wider horizontal strip.

Proof. If $|\Im z| < \sigma_*$, the extra Fourier factor has magnitude at most $\exp(2\pi |k| |\Im z|)$. The upper bound (6.5) makes the series locally normally convergent because $2\pi |\Im z| < \pi^2/L$. Hence it is holomorphic in $\mathcal{S}_*$.

Suppose it continued to $|\Im z| < \sigma_* + \delta$. Periodicity continues by the identity theorem. For $k > 0$, shift the Fourier-coefficient contour to the compact line $\Im z = -(\sigma_* + \delta/2)$. Boundedness on one period gives

$$|c_k| \leq C \exp(-2\pi(\sigma_* + \delta/2)k),$$

contradicting (6.4), because $2\pi\sigma_* = \alpha$. Shifting upward gives the same contradiction for negative modes. $\square$

Remark 6.3. Maximality of the symmetric strip does not by itself prove that every point of either boundary line is singular. The stronger natural-boundary statement is a conjecture in [section 11](#).

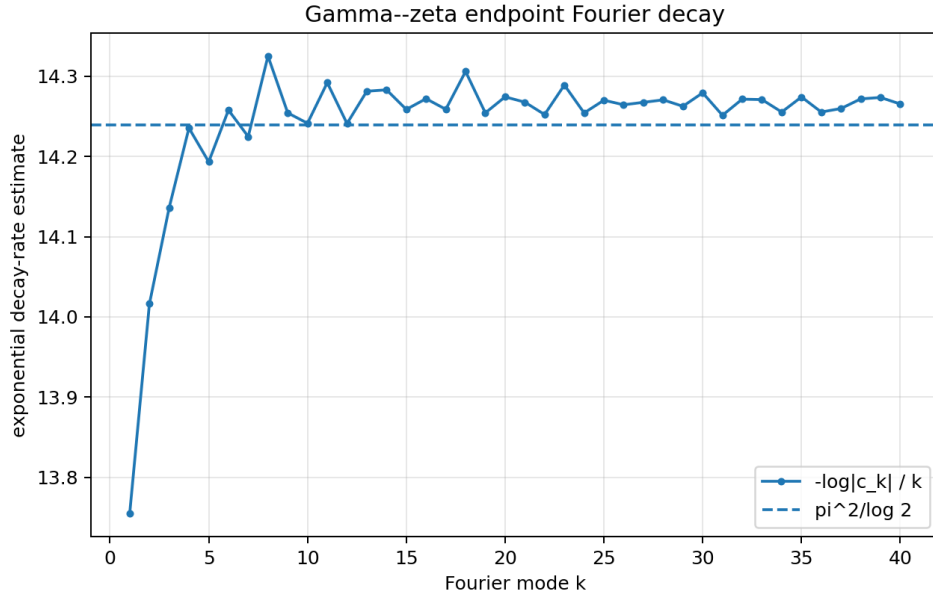


Figure 2: High-precision values of $-\log |c_k|/k$ and the exact limiting constant $\pi^2/\log 2$. Small oscillations come from the zeta factor; the Gamma factor fixes the exponential rate.

7 Weighted Wiener algebras and Bell-polynomial stability

The maximal strip gives quantitative control of every differential-polynomial periodic coefficient built from A_0 .

Definition 7.1. For $\sigma \geq 0$, let $\mathcal{A}_\sigma$ be the weighted Wiener algebra of one-periodic Fourier series

$$f(z) = \sum_{k \in \mathbb{Z}} \hat{f}(k) e^{2\pi i k z}, \quad \|f\|_\sigma = \sum_{k \in \mathbb{Z}} |\hat{f}(k)| e^{2\pi \sigma |k|} < \infty.$$

Theorem 7.2 (Wiener–Bell calculus). *Let $0 \leq \tau < \sigma$.*

1. $\mathcal{A}_\sigma$ is a Banach algebra and $\|fg\|_\sigma \leq \|f\|_\sigma \|g\|_\sigma$.
2. If $f \in \mathcal{A}_\sigma$, then for $r \geq 1$,

$$\|f^{(r)}\|_\tau \leq \left(\frac{r}{e(\sigma - \tau)} \right)^r \|f\|_\sigma. \quad (7.1)$$

3. If $S_K f = \sum_{|k| \leq K} \hat{f}(k) e^{2\pi i k z}$, then

$$\sup_{|\Im z| \leq \tau} |f(z) - S_K f(z)| \leq e^{-2\pi(\sigma - \tau)(K+1)} \|f\|_\sigma. \quad (7.2)$$

4. For the complete exponential Bell polynomial $\mathcal{B}_n$ and $f_1, \dots, f_n \in \mathcal{A}_\sigma$,

$$\|\mathcal{B}_n(f_1, \dots, f_n)\|_\sigma \leq \mathcal{B}_n(\|f_1\|_\sigma, \dots, \|f_n\|_\sigma). \quad (7.3)$$

In particular, if $g = \exp f$, then

$$\|g^{(n)}\|_\tau \leq e^{\|f\|_\tau} \mathcal{B}_n(\|f'\|_\tau, \dots, \|f^{(n)}\|_\tau). \quad (7.4)$$

Proof. The product estimate is the weighted convolution inequality. For derivatives, the Fourier multiplier is $(2\pi i k)^r$, and

$$(2\pi x)^r e^{-2\pi(\sigma - \tau)x} \leq \left(\frac{r}{e(\sigma - \tau)} \right)^r.$$

The tail estimate follows by extracting the factor $e^{-2\pi(\sigma - \tau)(K+1)}$. Complete Bell polynomials have nonnegative scalar coefficients, so repeated use of the Banach-algebra inequality proves (7.3). The last assertion is Faà di Bruno's formula $g^{(n)} = g \mathcal{B}_n(f', \dots, f^{(n)})$ and $\|\exp f\|_\tau \leq \exp \|f\|_\tau$. $\square$

Corollary 7.3 (certified endpoint coefficient truncation). *The leading Fabius endpoint fluctuation A_0 belongs to $\mathcal{A}_\sigma$ for every $\sigma < \sigma_*$. Any coefficient obtained from finitely many derivatives, products, Bell polynomials, and scalar Bernoulli constants belongs to the same family of algebras. Its Fourier truncation error on every smaller strip is exponentially certified by (7.2).*

This supplies a functional-analytic justification for a practical operation repeatedly needed in the all-orders endpoint theory: compute a short Fourier representation of the primitive periodic term, differentiate and combine it through Bell polynomials, and retain rigorous control rather than relying only on pointwise numerical smallness.

8 Radial–angular Lambert phase tomography

The multiplicative phase orbits recover values of periodic coefficient functions. A phase grid then recovers their Fourier spectra.

8.1 Radial interpolation of the coefficient jet

Fix a rational phase θ , an admissible ρ , $q = \rho^{-1}$, and $H = (n + \theta)^{-1}$. Suppose

$$y_s = \mathcal{E}(\rho^s(n + \theta)) = \sum_{j=0}^p B_j q^{sj} + r_s, \quad B_j = A_j(\theta) H^j. \quad (8.1)$$

Let $\ell_s(t)$ be the Lagrange basis at the geometric nodes $1, q, \dots, q^p$:

$$\ell_s(t) = \prod_{\substack{0 \leq r \leq p \\ r \neq s}} \frac{t - q^r}{q^s - q^r}. \quad (8.2)$$

Define

$$\hat{A}_j(\theta) = H^{-j} [t^j] \sum_{s=0}^p y_s \ell_s(t). \quad (8.3)$$

Theorem 8.1 (radial jet recovery). *If $r_s = 0$, (8.3) recovers $A_0(\theta), \dots, A_p(\theta)$ exactly. In general,*

$$|\hat{A}_j(\theta) - A_j(\theta)| \leq H^{-j} \sum_{s=0}^p |[t^j] \ell_s(t)| |r_s|. \quad (8.4)$$

If $r_s = \mathcal{O}(H^{p+1} q^{s(p+1)})$ uniformly in phase, then

$$\hat{A}_j(\theta) - A_j(\theta) = \mathcal{O}(H^{p+1-j}). \quad (8.5)$$

For $j = 0$, the weights are exactly (3.10).

Proof. The sample vector without remainders is the evaluation of the polynomial $\sum_{j=0}^p B_j t^j$ at $t = q^s$. Lagrange interpolation reconstructs it exactly. Subtracting the remainder interpolant gives (8.4). Since p and q are fixed, the coefficient weights are fixed constants, proving (8.5). $\square$

The full coefficient weights also have a closed q -binomial form. Put $X_p = \{1, q, \dots, q^p\}$. The denominator is

$$\prod_{r \neq s} (q^s - q^r) = (-1)^s q^{sp-s(s+1)/2} (q; q)_s (q; q)_{p-s}, \quad (8.6)$$

and

$$[t^j] \ell_s(t) = \frac{(-1)^{p-j} e_{p-j}(X_p \setminus \{q^s\})}{(-1)^s q^{sp-s(s+1)/2} (q; q)_s (q; q)_{p-s}}. \quad (8.7)$$

The excluded elementary symmetric polynomial can be reduced to Gaussian binomials through

$$e_m(X_p \setminus \{q^s\}) = \sum_{a=0}^m (-q^s)^a q^{(m-a)(m-a-1)/2} \begin{bmatrix} p+1 \\ m-a \end{bmatrix}_q. \quad (8.8)$$

Thus the entire radial coefficient jet is governed by the same q -binomial algebra as the zeroth-order Richardson row.

8.2 Angular DFT and exact aliasing

Let a one-periodic coefficient have an absolutely convergent Fourier expansion

$$A(u) = \sum_{k \in \mathbb{Z}} c_k e^{2\pi i k u}. \quad (8.9)$$

Sample it at the rational phases $u_r = r/d$ and define

$$\tilde{c}_k^{(d)} = \frac{1}{d} \sum_{r=0}^{d-1} A(r/d) e^{-2\pi i k r/d}. \quad (8.10)$$

Theorem 8.2 (exact phase alias identity). *For every integer k ,*

$$\tilde{c}_k^{(d)} = \sum_{m \in \mathbb{Z}} c_{k+md}. \quad (8.11)$$

If, for some $\beta > 0$, $|c_k| \leq C e^{-\beta|k|}$ and $|k| \leq (d-1)/2$, then

$$|\tilde{c}_k^{(d)} - c_k| \leq \frac{2C e^{-\beta(d-|k|)}}{1 - e^{-\beta d}}. \quad (8.12)$$

Proof. Insert (8.9) into (8.10). The finite geometric sum is one precisely when the Fourier index is congruent to k modulo d , and zero otherwise. For the bound, every nonprincipal alias satisfies $|k+md| \geq |m|d - |k|$; sum the two geometric tails. $\square$

By [theorem 6.1](#), for every $\varepsilon > 0$ the Fabius coefficients obey $|c_k| \leq C_\varepsilon \exp[-(\alpha - \varepsilon)|k|]$. Hence the DFT alias error has the sharp available exponential scale $\alpha = \pi^2/L$.

8.3 The combined tomography theorem

Theorem 8.3 (two-axis Lambert phase tomography). *Let $d \geq 2$, choose $\rho \equiv 1 \pmod{d}$, and use the radial order- p estimator at each phase r/d . Suppose the normalized endpoint residual has the uniform expansion (2.4) through order $p+1$, and suppose $A_0 \in \mathcal{A}_\sigma$ for some $\sigma > 0$. For $|k| \leq (d-1)/2$, the DFT of the radial estimates satisfies*

$$\hat{c}_k^{(d,p,n)} - c_k = \mathcal{O}(n^{-p-1}) + \mathcal{O}\left(e^{-2\pi\sigma(d-|k|)}\right), \quad (8.13)$$

uniformly over the phase grid. For the leading Fabius periodic term one may take every $\sigma < \pi/(2L)$, so the angular exponent can be made arbitrarily close to π^2/L .

Proof. Equation (5.6) gives a uniform $\mathcal{O}(n^{-p-1})$ error at each phase. Averaging by a DFT does not increase the maximum absolute error. Apply [theorem 8.2](#) to the exact phase values, with the Fourier bound supplied by membership in $\mathcal{A}_\sigma$. $\square$

The same construction applies to every A_j : use the full radial jet estimator (8.3), then a DFT in the phase variable. By [theorem 7.3](#), periodic coefficients formed through the audited Bell/Bernoulli saddle calculus inherit exponentially controlled phase spectra.

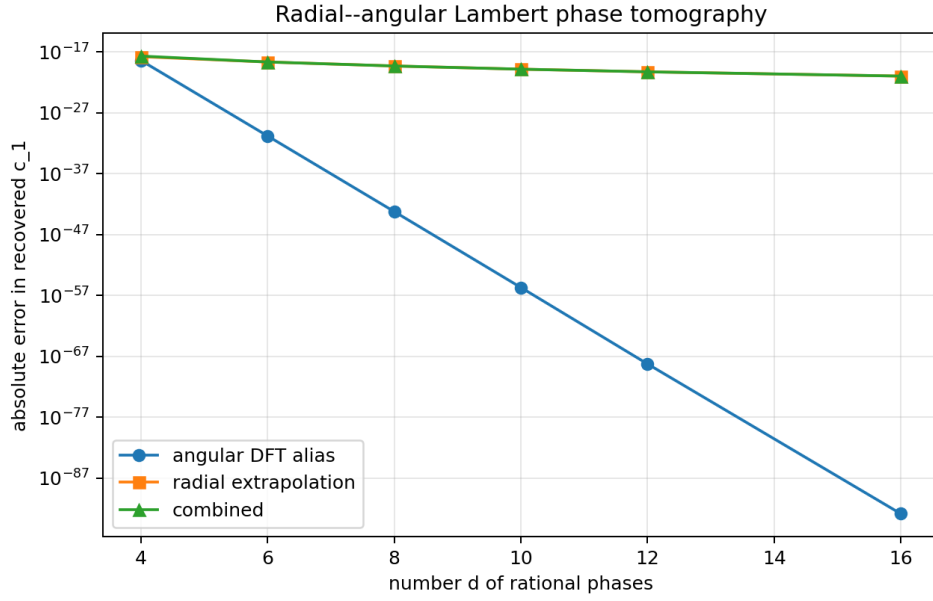


Figure 3: A controlled two-axis experiment recovering c_1 . The angular samples use the actual Gamma–zeta periodic term; documented smooth surrogate functions supply higher inverse-Lambert coefficients so that the radial cancellation can be tested independently.

9 Numerical and exact-arithmetic experiments

The accompanying `numerical_experiments.py` is deliberately independent of the LaTeX source. It uses exact `Fraction` arithmetic for all polynomial filters and 90-decimal-digit `mpmath` arithmetic for Gamma–zeta calculations. The zeta function is forced to use Euler–Maclaurin evaluation rather than an automatic method selection, avoiding threshold-dependent loss of reproducibility on the line $\Re s = 1$.

9.1 Exact filter verification

For $q = 1/2$ and every order $1 \leq m \leq 10$, the script checks

$$\sum_j w_j(N+j)^\ell q^{N+j} = 0, \quad 0 \leq \ell < m,$$

for both the minimal and twisted Thue–Morse filters. Every residual is an exact rational zero. It also verifies the mixed filter (3.8).

Table 2: Span–conditioning tradeoff at $q = 1/2$.

m	minimal span	minimal κ	TM span	TM κ
1	1	3	1	3.0000000000
2	2	9	3	5.0000000000
3	3	27	7	5.6666666667
4	4	81	15	5.7111111111
5	5	243	31	5.7112854031
6	6	729	63	5.7112854057
8	8	6561	255	5.7112854057
10	10	59049	1023	5.7112854057

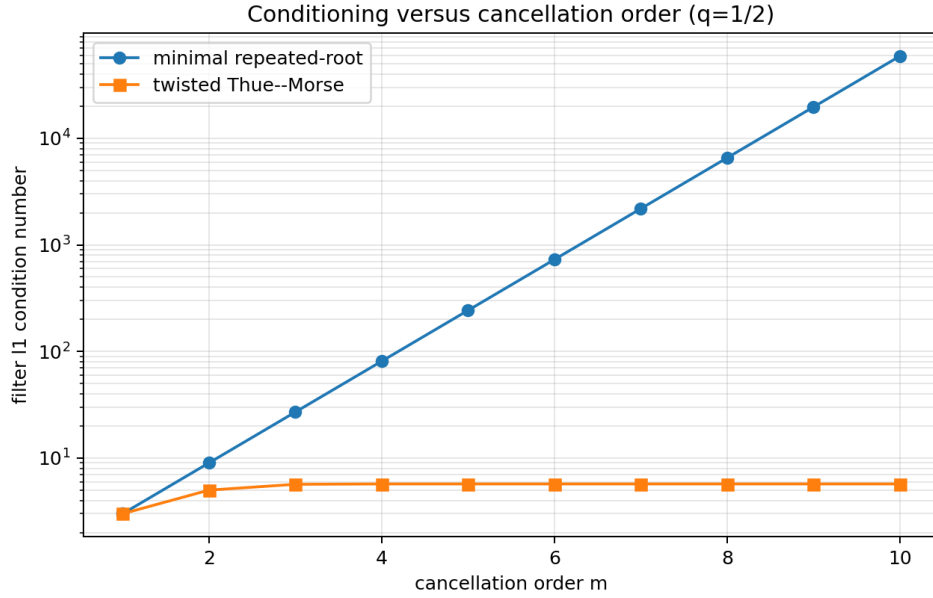


Figure 4: Minimal repeated-root and twisted Thue–Morse condition numbers. The vertical axis is logarithmic; the digital curve rapidly approaches its finite limit.

9.2 Gamma–zeta modes

The first nonzero modes are:

Table 3: High-precision magnitudes of the endpoint Fourier coefficients.

k	$ c_k $	$-\log c_k /k$
1	$1.06271162518632 \times 10^{-6}$	13.7546867793
2	$6.69278636792434 \times 10^{-13}$	14.0162879620
3	$3.82107872358669 \times 10^{-19}$	14.1361946653
4	$1.86460518002229 \times 10^{-25}$	14.2353944985
5	$1.51268570169932 \times 10^{-31}$	14.1932502403
10	$1.41459209881527 \times 10^{-62}$	14.2413434545

The exact limiting rate is

$$\alpha = 14.2388293249875054338606757113\dots, \quad \sigma_* = 2.26618007091359690481384147286\dots$$

9.3 Tomography experiment

The angular part evaluates the actual truncated Gamma–zeta series for A_0 . To test radial cancellation without silently importing unparsed formulas for the higher audited saddle coefficients, the program defines explicit smooth surrogate coefficients

$$\hat{A}_j(k) = 0.2^j \left(1 + \frac{k^2}{(j+1)^2} \right) c_k, \quad j \geq 1.$$

This surrogate has the same real symmetry and strip structure; it is not claimed to equal the true A_j . Radial order $p = 3$, starting integer $n = 16$, and $\rho = d + 1$ give:

Table 4: Recovery error for c_1 in the controlled radial–angular model.

d	angular alias	radial error	combined error
4	3.8211×10^{-19}	1.6041×10^{-18}	1.9862×10^{-18}
6	1.5127×10^{-31}	2.1303×10^{-19}	2.1303×10^{-19}
8	5.7018×10^{-44}	4.7054×10^{-20}	4.7054×10^{-20}
10	1.9284×10^{-56}	1.4089×10^{-20}	1.4089×10^{-20}
12	5.2972×10^{-69}	5.1635×10^{-21}	5.1635×10^{-21}
16	1.2993×10^{-93}	1.0305×10^{-21}	1.0305×10^{-21}

The exact rational phase drift is zero in every sample. Once $d \geq 6$, angular aliasing is negligible relative to the order-three radial remainder, exactly as (8.13) predicts.

10 Consequences for the Fabius–Rvachev research program

10.1 A single annihilator language

Three constructions that look unrelated become instances of one spectral polynomial.

1. Ordinary Richardson extrapolation puts simple roots at the known error ratios.
2. Confluent Richardson puts repeated roots at ratios carrying polynomial prefactors.
3. Prouhet–Thue–Morse cancellation is a digitally factorized repeated root, with the exponential twist chosen so that the geometric factor cancels before the Prouhet sum is taken.

This shift-polynomial language is especially convenient for Lean: the analytic application is reduced to a finite polynomial divisibility lemma plus a norm estimate.

10.2 Endpoint coefficient spectroscopy

The exact-Lambert phase is not only an asymptotic bookkeeping device. Rational multiplicative orbits turn the endpoint into a laboratory in which one may recover the periodic coefficient functions A_j separately. A practical high-precision workflow is:

1. choose a phase denominator d and $\rho \equiv 1 \pmod{d}$;
2. evaluate the normalized log Fabius residual at $x_{r,s} = \lambda_{r,s} 2^{-\lambda_{r,s}}$, where $\lambda_{r,s} = \rho^s(n + r/d)$;
3. perform radial q -interpolation to recover $A_j(r/d)$;
4. perform a DFT over r to recover Fourier coefficients;
5. certify radial error from the next asymptotic order and angular error from the Wiener strip.

The method applies equally to coefficients in the inverse-Fabius endpoint expansion once the corresponding normalized residual is written in the exact Lambert coordinate.

10.3 Bell and Bernoulli closure

The corpus’s Bell-polynomial coefficient calculus often makes formulas look combinatorially explosive. In the weighted Wiener algebra, however, every product and Bell composition is norm-submultiplicative. Bernoulli numbers enter as scalar coefficients, while derivatives are controlled by the strip margin. This suggests a certified implementation architecture: store periodic functions as weighted Fourier vectors, build each all-orders coefficient by Bell operations, and propagate a scalar norm bound in parallel. Numerical truncation then comes with a theorem-level remainder certificate.

10.4 Generalized Pascal–Rvachev hierarchies

Higher-rank Mellin poles naturally produce polynomial factors in the logarithmic scale. Along a multiplicative phase orbit these are exactly the $s^\ell q^{rs}$ modes treated by confluence. The digital filter is therefore likely to be more useful for the hierarchy than for the classical rank-one case: higher pole orders increase confluence, and

the minimal filter's condition number grows exponentially in that order, while a digital factorization can keep it bounded.

11 Conjectures and promising research directions

Conjecture 1: natural boundary of the endpoint fluctuation. The two lines

$$\Im z = \pm \frac{\pi}{2 \log 2}$$

are natural boundaries for the leading periodic endpoint function A_0 . The maximal-strip theorem proves that some obstruction occurs at each side, but does not locate the singular set. A route to proof is to analyze the phase of $\Gamma(-\chi_k)\zeta(1-\chi_k)$ and apply a gap-free natural-boundary criterion for Fourier series of exact exponential type.

Conjecture 2: sharp DFT alias exponent. For every fixed nonzero mode k , the exact phase-grid alias $\sum_{m \neq 0} c_{k+md}$ satisfies

$$\log \left| \sum_{m \neq 0} c_{k+md} \right| = -\frac{\pi^2}{\log 2} d + o(d)$$

outside, at most, a sparse exceptional set of grid sizes. The upper exponential rate is proved here; the issue is ruling out systematic phase cancellation among the nearest aliases.

Open problem 11.1 (optimal span–conditioning frontier). For $q \in (0, 1)$ define

$$K_m(q; D) = \inf \left\{ \sum_{j=0}^D |w_j| : W(1) = 1, (z - q)^m \mid W(z), \deg W \leq D \right\}.$$

Determine $K_m(q; D)$ uniformly between the endpoints

$$K_m(q; m) = \left(\frac{1+q}{1-q} \right)^m, \quad \inf_D K_m(q; D) = 1.$$

The Prouhet factorizations provide explicit upper bounds. The binary choice is uniquely minimal-span among collision-free complete digital stencils, but the full Pareto frontier appears to be a nontrivial extremal problem in polynomial approximation and linear programming.

Open problem 11.2 (irrational near-locks). For irrational phase θ , exact multiplicative locks do not exist. Continued-fraction approximants can choose ρ with $\|(\rho - 1)\theta\|$ small, but the phase defect is multiplied by $1 + \rho + \dots + \rho^{s-1}$. Develop a simultaneous Diophantine criterion that balances this phase drift against the bounded Richardson weights and the strip-controlled phase derivative. The expected result is a provable work–accuracy tradeoff for irrational phase tomography.

Conjecture 3: common strip in the Pascal–Rvachev hierarchy. After subtraction of the rank-dependent logarithmic polynomial, every periodic coefficient in the Pascal–Rvachev endpoint hierarchy has maximal strip half-width $\pi/(2 \log 2)$. Higher rank changes only polynomial factors in the Fourier mode and hence the boundary singularity order, not the Gamma-controlled exponential type.

Conjecture 4: boundary singularities control late Bell coefficients. The large-order growth of the all-orders endpoint and inverse-endpoint Bell-polynomial coefficients is governed by the nearest singularities on the two strip boundaries. After appropriate normalization, the coefficient sequence should admit a resurgent factorial-over-power law with phase determined by the boundary singularities. Establishing this would connect the exact Gamma–zeta Fourier spectrum to optimal truncation and exponentially improved Lambert asymptotics.

11.1 A concrete Lean formalization order

The new results break naturally into short formal modules.

1. Define polynomial shift operators on sequences valued in a normed additive group.
2. Prove [theorem 3.1](#) for scalar geometric-polynomial modes.
3. Formalize root multiplicity, the minimal annihilator, and the finite norm bound.
4. Import the existing finite Thue–Morse polynomial and prove $\mathcal{T}_{m,q} = T_m(z/q)/T_m(1/q)$, coefficient formula (4.6), and norm formula (4.9).
5. Prove the modular arithmetic phase lock [theorem 5.1](#) over rationals.
6. Reuse the existing Gaussian q -binomial Lagrange development for the radial jet.
7. Formalize the finite DFT alias identity; postpone the complex-analytic maximal-strip theorem until Gamma and zeta estimates are available in the chosen library layer.

The first six items are predominantly finite algebra and should be substantially easier than the endpoint analytic inputs already present in the project.

12 Conclusion

The main structural discovery is not another isolated formula for F , F^{-1} , or $\widehat{\text{up}}$. It is a reusable spectral calculus.

A root of a shift polynomial represents an asymptotic geometric mode; multiplicity represents its polynomial decoration. The standard q -Richardson row uses simple roots. Confluence handles logarithmic and polynomial decorations. A Thue–Morse product spreads a repeated root over a binary stencil, exchanging span for bounded conditioning. The exact Lambert coordinate turns rational endpoint phases into geometric orbits on which those filters act. Finally, the Gamma–zeta mode formula supplies the sharp analytic strip needed to reconstruct the phase dependence by DFT and to certify every subsequent Bell-polynomial operation.

This package gives both theorem-level results and a practical experimental protocol. It also identifies clear next targets: the span-conditioning extremal problem, natural boundaries, irrational phase locks, hierarchy-wide strip theorems, and resurgent late-order analysis. Each direction is tightly connected to existing formalized or audited machinery, while avoiding duplication of the substantial frontier results already present in the repository.

A Auxiliary identities

A.1 Complete homogeneous form of transformed moments

For the geometric Lagrange weights, let

$$\mu_{p,r}(q) = \sum_{s=0}^p \omega_{p,s}(q) q^{sr}.$$

Then $\mu_{p,0} = 1$, $\mu_{p,r} = 0$ for $1 \leq r \leq p$, and for $r \geq p+1$,

$$\mu_{p,r}(q) = (-1)^p q^{p(p+1)/2} h_{r-p-1}(1, q, \dots, q^p), \quad (\text{A.1})$$

where h_j is the complete homogeneous symmetric polynomial. The identity follows from the remainder of interpolation of t^r at $1, q, \dots, q^p$. The standard specialization

$$h_{r-p-1}(1, q, \dots, q^p) = \left[\begin{matrix} r-1 \\ p \end{matrix} \right]_q$$

recovers (3.11).

A.2 Additive phase weights

For completeness, the derivation of (5.7) is one line. At nodes $h_j = (\lambda + j)^{-1}$, the Lagrange weight at zero is

$$\prod_{r \neq j} \frac{-h_r}{h_j - h_r} = \prod_{r \neq j} \frac{-(\lambda + j)}{r - j} = \frac{(-1)^{p-j} (\lambda + j)^p}{j!(p-j)!}.$$

This exact formula makes the conditioning contrast transparent; it is not a numerical artifact of solving a Vandermonde system.

A.3 Digital condition limit

The finite identity

$$\prod_{j=0}^{m-1} (1 + q^{2^j}) = \frac{1 - q^{2^m}}{1 - q}$$

is simply the binary factorization of $1 - q^{2^m}$. Dividing by $\Theta_m(q)$ gives (4.9); monotone convergence of $\Theta_m(q)$ gives (4.10).

B Reproduction guide

From the extracted report directory, run

```
python numerical_experiments.py --output-dir . --precision 90
latexmk -pdf -interaction=nonstopmode -halt-on-error fabius_frontier_report.tex
```

The script regenerates:

- filter_conditioning.csv and conditioning_tradeoff.png;
- lambert_conditioning.csv and lambert_conditioning.png;
- gamma_zeta_coefficients.csv and gamma_zeta_decay.png;
- tomography_errors.csv and tomography_errors.png;
- experiment_results.txt and verification.json.

Exact filter checks use Python’s Fraction; no tolerance is involved. Floating experiments use mpmath. The source comments document the surrogate higher endpoint coefficients and every normalization.

C Complete pinned LaTeX inventory

The following manifest is included as a separate plain-text file in the archive. It records the complete recursive 78-path corpus boundary used for the nonduplication audit.

```
Pinned repository snapshot
=====
Repository: VladimirReshetnikov/ProveIt
Commit: f49aad5c9c6d627a64822ea9f8e46f5270b95cef
Documentation subtree: Analysis/FabiusFunction/docs
Recursive docs tree SHA: 6063d0d90cd9a4d25ca74cd4f8b80448313fce64
Inventory method: recursive Git tree, not GitHub’s lagging code-search index
LaTeX paths in the pinned subtree: 78

The inventory includes ordinary manuscripts, historical revisions, imported/corrected
source papers, and six small generated TeX table/check fragments. Historical copies were
read as lineages and checked against their current successors rather than counted as
independent mathematical evidence.

Paths
=====
```

1. FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
2. Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
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